

Topic 5.2

Sigma Notation & Limits of Finite Sums

* Definition of Sigma Notation: The sigma notation

for the finite sum $a_1 + a_2 + \dots + a_n$ is denoted

by $\sum_{k=1}^n a_k$
index $k=1$ lower limit

→ Used to write lengthy sums in compact form
→ Sigma (Σ) is an upper case letter in Greek
→ Σ is called Sigma/Summation

a_1 is the first term, a_2 is the second term, $\dots$, a_k is the k th term.

• Example 1: Evaluate the value of the following summations:

(a) $\sum_{k=1}^5 k^2$ (b) $\sum_{k=2}^6 (k-1)$ (c) $\sum_{k=1}^4 (-1)^k \cos(k\pi)$

Solution (a) $\sum_{k=1}^5 k^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2 = \boxed{55}$ Ans

(b) $\sum_{k=2}^6 (k-1) = (2-1) + (3-1) + (4-1) + (5-1) + (6-1) = \boxed{15}$ Ans

(c) $\sum_{k=1}^4 (-1)^k \cos k\pi = (-1)^1 \cos \pi + (-1)^2 \cos 2\pi + (-1)^3 \cos 3\pi + (-1)^4 \cos 4\pi = \boxed{4}$ Ans
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Note

A sum can be written in more than one notations if we change the limits

① $\sum_{k=1}^5 2k = 2+4+6+8+10$

② $\sum_{k=0}^4 (2k+2) = 2+4+6+8+10$

- Example 2: Express the following sums in Sigma notation.

(a) $1+3+5+7+9+11+13 = ?$

Solution: $1+3+5+7+9+11+13 = \sum_{k=1}^7 (2k-1)$ Ans

$= \sum_{k=10}^{17} (2k-19)$ Ans

where k starts from any integer.

(b) $-\frac{1}{5} + \frac{2}{5} - \frac{3}{5} + \frac{4}{5} - \frac{5}{5} + \frac{6}{5}$

Solution: $-\frac{1}{5} + \frac{2}{5} - \frac{3}{5} + \frac{4}{5} - \frac{5}{5} + \frac{6}{5} = \sum_{k=1}^6 \frac{(-1)^k k}{5}$ Ans

* Algebra Rules for finite sums:

① $\sum_{k=1}^n (a_k \pm b_k) = \sum_{k=1}^n a_k \pm \sum_{k=1}^n b_k$

② $\sum_{k=1}^n C a_k = C \sum_{k=1}^n a_k$

③ $\sum_{k=1}^n C = n \cdot C$, where 'C' is constant

→ If upper & lower limits are same then the summation reduces to just one term.

$\sum_{k=2}^2 k^2 = 2^2 = 4$

→ If in the summation, expression on right of summation does not involve the index then

$\sum_{i=1}^5 2 = 2+2+2+2+2 = 10$

$\sum_{k=3}^6 x^3 = x^3 + x^3 + x^3 + x^3$

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- Example 1: Find the value of $\sum_{k=1}^7 \pi$.

Solution:

$$\sum_{k=1}^7 \pi = \boxed{7\pi} \quad \underline{\text{Ans}}$$

- Example 2: Suppose that $\sum_{k=1}^n a_k = -5$ and $\sum_{k=1}^n b_k = 6$. Find the value of $\sum_{k=1}^n (b_k - 2a_k)$.

Solution: $\sum_{k=1}^n b_k - \sum_{k=1}^n 2a_k = \sum_{k=1}^n b_k - 2 \sum_{k=1}^n a_k$

$$= 6 - 2(-5) = 6 + 10 = \boxed{16} \quad \underline{\text{Ans}}$$

* Some formulas for positive integers:

If 'n' is a positive integer, then

① $\sum_{k=1}^n k = \frac{n(n+1)}{2}$

④ $\sum_{k=1}^n 1 = n$

② $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

③ $\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2$

- Example 1: Evaluate the following sums:

(a) $\sum_{k=1}^7 k(2k+1)$ (b) $\left(\sum_{k=1}^7 k \right)^2 - \sum_{k=1}^7 \frac{k^3}{4}$

Solution: (a) $\sum_{k=1}^7 (2k^2 + k) = 2 \sum_{k=1}^7 k^2 + \sum_{k=1}^7 k$

$$= 2 \left[\frac{7(7+1)(14+1)}{6} \right] + \frac{7(7+1)}{2}$$

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$$\sum_{k=1}^7 (2k^2 + k) = \frac{7 \cdot 8 \cdot 15}{3} + \frac{7 \cdot 8^2}{2} = (7 \cdot 8 \cdot 5) + (7 \cdot 4) = 280 + 28 = 308 \quad \text{Ans}$$

$$\begin{aligned} \text{(b)} \left(\sum_{k=1}^7 k \right)^2 - \sum_{k=1}^7 k^3 &= \left(\frac{7(7+1)}{2} \right)^2 - \frac{1}{4} \sum_{k=1}^7 k^3 \\ &= \frac{49 \cdot 64}{4} - \frac{1}{4} \left(\frac{7^2(7+1)^2}{2^2} \right) \\ &= (49 \cdot 16) - (49 \cdot 16) = 0 \quad \text{Ans} \end{aligned}$$

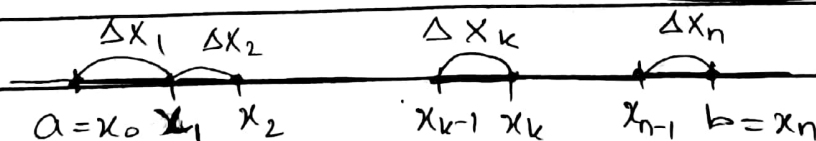
* Definition of Partition: Let $x_1, x_2, \dots, x_{n-1}$ be the points between 'a' & 'b' such that

$$x_0 = a < x_1 < x_2 < \dots < x_{n-1} < x_n = b.$$

① The set $P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$ is called a partition of $[a, b]$.

② The length of the k th subinterval $[x_{k-1}, x_k]$ is $\Delta x_k = x_k - x_{k-1}$.

$$\boxed{\text{Length of sub-interval} = \Delta x_k = x_k - x_{k-1}}$$



Δx_k can be different (It may have equal width and it may have unequal width if no. of rectangles are not mentioned)

• Example 1: Consider the interval $[2, 7]$,
let $P = \{2, 3, 4.5, 5, 7\}$. Find Δx_k (width).

Solution: $\Delta x_1 = x_1 - x_0 = 3 - 2 = 1$

$$\Delta x_2 = x_2 - x_1 = 4.5 - 3 = 1.5$$

$$\Delta x_3 = x_3 - x_2 = 5 - 4.5 = 0.5$$

$$\Delta x_4 = x_4 - x_3 = 7 - 5 = 2$$

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* Definition of Riemann Sum: Let f be a continuous function on $[a, b]$, let $P = \{x_0, x_1, \dots, x_{n-1}, x_n\}$ be a partition of $[a, b]$ and let $c_k \in [x_{k-1}, x_k]$. Then a Riemann sum for the area enclosed by the function ' f ' on $[a, b]$ is defined as

Imp * Area As Limit:

$$S_p = \sum_{k=1}^n f(c_k) \Delta x_k$$

$$\text{Area} = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x_k$$

* Riemann Sum:

$$S_p = \sum_{k=1}^n f(c_k) \Delta x_k$$

Riemann Sum = $S_p = f(c_1) \Delta x_1 + f(c_2) \Delta x_2 + \dots + f(c_n) \Delta x_n$

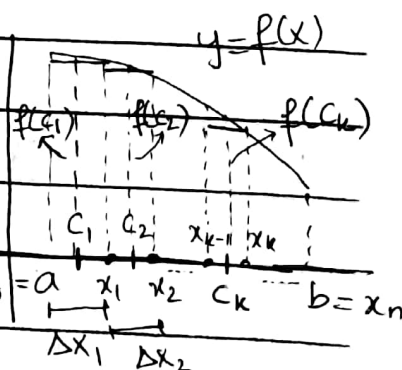
* Definition of Norm of a Partition: The norm of a partition $P = \{x_0, x_1, \dots, x_{n-1}, x_n\}$ is

The norm of a partition

$P = \{x_0, x_1, \dots, x_{n-1}, x_n\}$ is

length of 1st subinterval.

length of sub-intervals of partition P is called norm of partition.



Imp $\|P\| = \max \{\Delta x_1, \Delta x_2, \dots, \Delta x_n\}$

To find c_k : ① Left Endpoint: $c_k = x_{k-1} = a + (k-1)\Delta x$

② Right Endpoint: $c_k = x_k = a + k\Delta x$

③ Midpoint: $c_k = \frac{1}{2}(x_{k-1} + x_k) = a + (k - \frac{1}{2})\Delta x$

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- Example 1: Find the norm of the partition

$$P = \{-2, -1.6, -0.5, 0, 0.8, 1\}$$

Solution: $\Delta x_1 = -1.6 - (-2) = 0.4$

$$\Delta x_2 = -0.5 - (-1.6) = 1.1$$

$$\Delta x_3 = 0 - (-0.5) = 0.5$$

$$\Delta x_4 = 0.8 - 0 = 0.8$$

$$\Delta x_5 = 1 - 0.8 = 0.2$$

So $\|P\| = \max\{0.4, 1.1, 0.5, 0.8, 0.2\}$

$$\|P\| = 1.1 \quad \text{Ans}$$



- Example 2: Let $f(x) = x$ over the interval $[1, 2]$.

(a) Find a formula for the Riemann sum obtained by dividing the interval $[1, 2]$ into n equal subintervals and using the right endpoint for each

Ck:

(b) Take a limit of the sum as $n \rightarrow \infty$ to calculate the area under the curve on $[1, 2]$.

Solution (a) $\Delta x = \frac{2-1}{n} = \frac{1}{n}$

For right endpoint

$$C_k = a + k \Delta x$$

Here $a = 1$ & $\Delta x = \frac{1}{n}$

$$C_k = 1 + \frac{k}{n}$$

• Δx is used when rectangles are of equal width

• Δx_k is used when rectangles are of unequal width

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$$f(c_k) \Delta x = c_k \cdot \Delta x \\ = \left(1 + \frac{k}{n}\right) \left(\frac{1}{n}\right)$$

$$f(c_k) \Delta x = \frac{1}{n} + \frac{k}{n^2}$$

$$\text{Riemann Sum} = S_p = \sum_{k=1}^n f(c_k) \Delta x$$

$$= \sum_{k=1}^n \left(\frac{1}{n} + \frac{k}{n^2} \right)$$

$$= \sum_{k=1}^n \frac{1}{n} + \sum_{k=1}^n \frac{k}{n^2}$$

$$= \frac{1}{n} \sum_{k=1}^n 1 + \frac{1}{n^2} \sum_{k=1}^n k$$

$$= \frac{1}{n} (n) + \frac{1}{n^2} \cdot \frac{n(n+1)}{2}$$

$$= 1 + \frac{n+1}{2n}$$

$$= 1 + \frac{n}{2n} + \frac{1}{2n}$$

$$= 1 + \frac{1}{2} + \frac{1}{2n} = \boxed{\frac{3}{2} + \frac{1}{2n}} \quad \text{Ans}$$

← →

$$(b) \text{ Area} = \lim_{n \rightarrow \infty} S_p = \lim_{n \rightarrow \infty} f(c_k) \Delta x$$

$$= \lim_{n \rightarrow \infty} \left(\frac{3}{2} + \frac{1}{2n} \right) = \frac{3}{2} + \frac{1}{\infty}$$

$$= \boxed{\frac{3}{2}} \quad \text{Ans}$$

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- Example 3: Same problem as before (Example 2) but with left endpoints.

Solution: (a) $\Delta x = \frac{b-a}{n} = \frac{2-1}{n} = \frac{1}{n}$.

left endpoint: $c_k = x_{k-1} = a + (k-1)\Delta x$.

Here $a=1$, $\Delta x = \frac{1}{n}$

$$c_k = 1 + (k-1)\left(\frac{1}{n}\right)$$

$$= 1 + \frac{k}{n} - \frac{1}{n} = 1 + \frac{k-1}{n}$$

Now

$$f(c_k) \Delta x = c_k \cdot \Delta x$$

$$= \left(1 + \frac{k-1}{n}\right) \frac{1}{n}$$

$$= \frac{1}{n} + \frac{k-1}{n^2}$$

$$\text{Riemann Sum} = S_p = \sum_{k=1}^n f(c_k) \Delta x.$$

$$= \sum_{k=1}^n \left(\frac{1}{n} + \frac{k-1}{n^2} \right)$$

$$= \sum_{k=1}^n \frac{1}{n} + \sum_{k=1}^n \frac{k-1}{n^2}$$

$$= \frac{1}{n} \sum_{k=1}^n 1 + \frac{1}{n^2} \sum_{k=1}^n (k-1)$$

$$= \frac{1}{n} (n) + \frac{1}{n^2} \sum_{k=1}^n k - \frac{1}{n^2} \sum_{k=1}^n 1$$

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$$S_p = 1 + \left(\frac{1 \cdot n(n+1)}{2} \right) - \frac{1}{n^2} (n).$$

$$= 1 + \frac{n+1}{2n} - \frac{1}{n}.$$

$$= 1 + \frac{n}{2n} + \frac{1}{2n} - \frac{1}{n}$$

$$= 1 + \frac{1}{2} + \frac{1}{2n} - \frac{1}{n} = \frac{3}{2} + \frac{1-2}{2n}$$

$$S_p = \left[\frac{3}{2} - \frac{1}{2n} \right] \text{ Ans}$$

$$(b) \text{ Area} = \lim_{n \rightarrow \infty} \sum_{k=1}^n S_p = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x.$$

$$= \lim_{n \rightarrow \infty} \left(\frac{3}{2} - \frac{1}{2n} \right) = \frac{3}{2} - \frac{1}{2(\infty)} = \left[\frac{3}{2} \right] \text{ Ans}$$

* Practice Question Use area definition with right endpoints of each subinterval to find area under parabola $y = 9 - x^2$ over $[0, 3]$.